

Open Science in Practice: Collaborative Research with Git



Niklas Edvall, PhD
niklas.edvall@ki.se

K8F6106
HT2026



Andreas Gerhardsson, PhD
andreas.Gerhardsson@ki.se

Git branching

Andreas Gerhardsson

Your Git journey so far



Your Git journey ahead





QUANTUM FLUX
STABILITY:
87%



Remember:
measure
twice?)
😊

Don't
forget
snacks
😊

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$$
$$\Psi = A e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}$$



QUANTUM FLUX
STABILITY:
87%



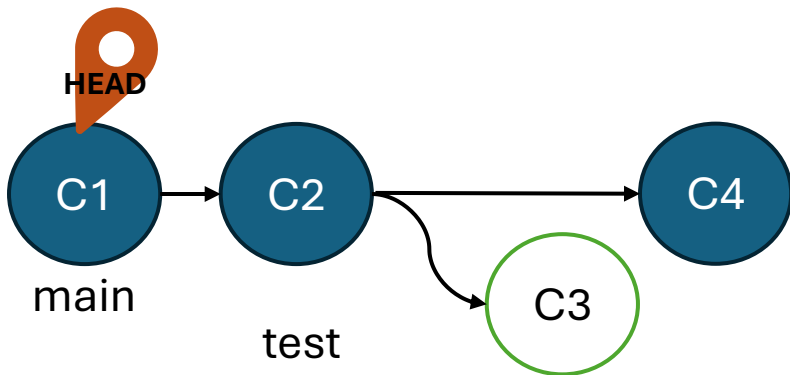
Don't
forget
snacks
😊

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$$
$$\Psi = A e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}$$

SCIENCE
IS
FUN

SCIENCE
IS
FUN

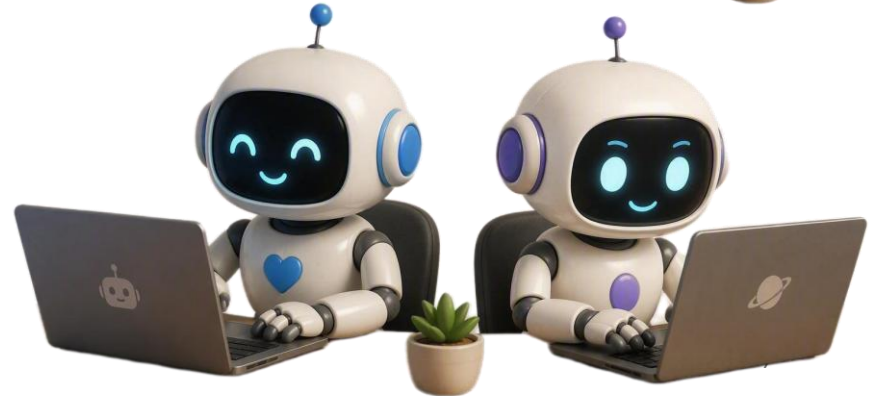
What is a branch?



```
andreas ~ % git commit -m "something"  
git switch -c test  
git commit -m "testing"  
git switch main  
git commit -m "something"
```

Why branching?

- **Safe experimentation**
- **Parallel work**
- **Collaboration**



Creating and switching branches

```
andreas ~ %
```

```
git status          # see status and what branch you are on

git branch           # list local branches

git branch --all     # list all branches

git switch <branch>  # switch branch

git switch -c <branch> # create new branch and switch to it

git log --all --oneline # show your git tree
```

Creating and switching branches



Project



read_data.R



run_models.R



make_figures.R



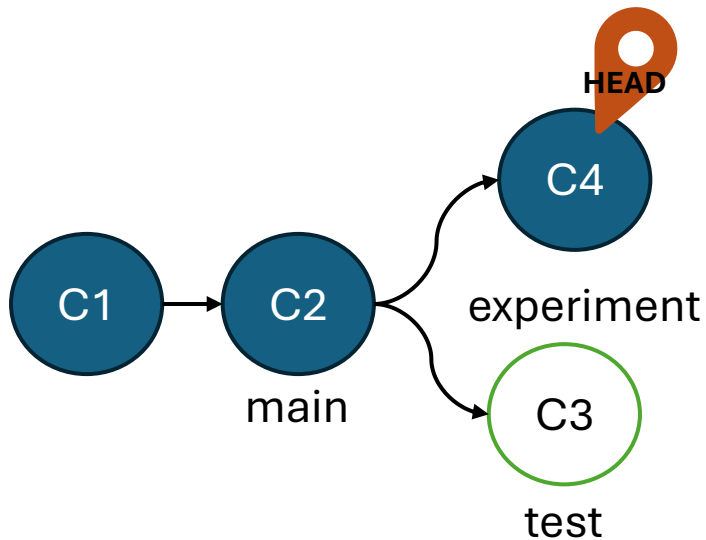
export.R



test.R

```
andreas ~ % git switch test
```

Merging a branch



Fast-forward merge

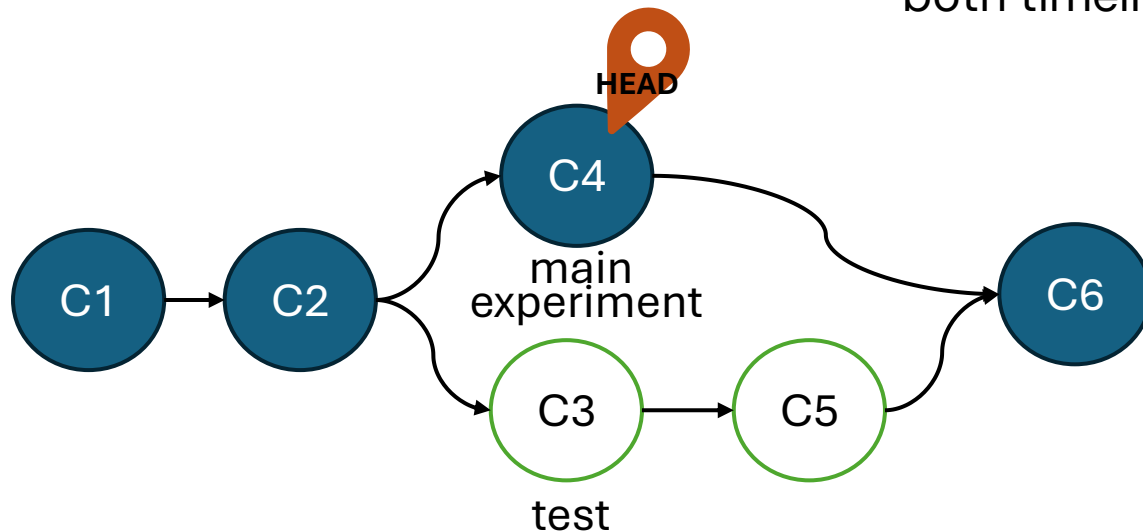
— the primary timeline stood still

```
andreas ~ % git switch main  
             git merge experiment
```

Merging a branch

Merge commit

— both timelines moved



```
andreas ~ % git switch main  
            git merge test
```

Merge conflicts



read_data.R (main)

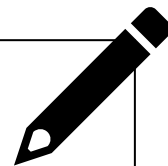
```
1 d <- read.data('data.csv')
2
3 mean(d$rain)
4
5
6
7
8
```



read_data.R (test)

```
1 d <- read.data('data.csv')
2
3 summary(d)
4
5
```

```
1 d <- read.data('data.csv')
2
3 summary(d)
4 mean(d$rain)
5
6
7
```



Renaming a branch

- Don't rename a branch if you suspect someone else is working on it
- Rename locally, and remote
- Delete old remote branch

```
andreas ~ % git branch --move test new-test  
git push --set-upstream origin new-test  
git push origin --delete test
```

Summary

- A branch is a movable pointer to a commit, not a copy of your files.
- Branching enables safe, isolated, parallel work on top of a trusted main.
- Merges join branches — either by fast-forward or by a merge commit.
- Conflicts are expected and resolvable; merge often to keep them small.



References

Chacon, S. (with Straub, B.). (2014). *Pro Git* (2nd ed). Apress L. P.